

The schematic illustrates the electrical connections between two connectors:

- J1 USB-C**: A USB Type-C connector with pins labeled A6 through B8.
- SMB1-SMB2**: Surface-mount components connected to pins B7 and B8.
- U4 USBCLC-25C6**: A USB-C-to-USB-A adapter chip with pins 1 through 6.
- USB-DN**: A standard USB Type-A female connector with pins labeled USB_DN+, USB_DN-, USB_DP+, and USB_DP-.
- Power Connections**: VBUS and GND are shown as common power and ground rails across all components.
- Data Connections**:
 - A6 connects to USB_DN+.
 - A5 connects to USB_DN-.
 - B5 connects to USB_DP+.
 - B6 connects to USB_DP-.
- Internal Adapter Connections**: Inside the adapter chip U4, pin 1 connects to USB_DN+, pin 2 to USB_DN-, pin 3 to USB_DP+, and pin 4 to USB_DP-.

U1: ESP32-S3-WROOM-1U-N16R2

xY5

EN 1 IN 2 TXD0 30 RXD0 31 TXD1 32 RXD1 33 TXD2 34 RXD2 35 TXD3 36 RXD3 37 BTN_LEFT 38 BTN_RIGHT 39 BTN_SEL 40 LCD_SCK 41 LCD_CS 42 LED_DATA 43 LED_VCC 44 GND

D0 4 D1 5 D2 6 D3 7 D4 8 D5 9 D6 10 D7 11 IN0 12 IN1 13 IN2 14 IN3 15 LCD_A 16 LCD_D 17 I2C_SDA 18 I2C_SCL 19 TXD0 20 RXD0 21 TXD1 22 RXD1 23 TXD2 24 RXD2 25 TXD3 26 RXD3 27 BTN_LEFT 28 BTN_RIGHT 29 BTN_SEL 30 LCD_SCK 31 LCD_CS 32 LED_DATA 33 LED_VCC 34 GND

SPI_MISO 10 SPI_MOSI 11 AUX0 12 AUX1 13 AUX2 14 AUX3 15 TECH1 16 TECH2 17 TECH3 18 TXD2 19 RXD2 20 TXD3 21 RXD3 22 BTN_LEFT 23 BTN_RIGHT 24 BTN_SEL 25 LCD_SCK 26 LCD_CS 27 LED_DATA 28 LED_VCC 29 GND

The diagrams show the following connections:

- J5 DISPLAY:** Pin 1 to +5V, Pin 2 to GND, Pins 3-14 to a 14-pin connector.
- J6 NAV_SW:** Pin 1 to +5V, Pin 2 to GND, Pins 3-6 to a 6-pin connector.
- J7 AUX:** Pin 1 to +3V3, Pin 2 to GND, Pins 3-8 to an 8-pin connector.

[illegible][illegible]

The circuit diagram shows the following components and connections:

- WDT_HOLD Pin:** Connected to a pull-up resistor R7 (BAT54S) to +5V and a capacitor C8 (220k) to GND.
- WOL_HOLD Pin:** Connected to a pull-up resistor R6 (1M) to +5V and a capacitor C9 (100nF) to GND.
- SSR_P1 Pin:** Connected to a pull-up resistor R5 (10k) to +5V and a capacitor C7 (100nF) to GND.
- SSR_P2 Pin:** Connected to a pull-up resistor R4 (10k) to +5V and a capacitor C6 (100nF) to GND.
- WOL_DEFEAT Pin:** Connected to a pull-up resistor R3 (100k) to +5V and a capacitor C5 (100nF) to GND.

[illegible]

The three diagrams illustrate different ways to connect a 3V3 supply to a microcontroller's VDD pins (N1, N2, N3):

- Diagram 1:** A 3V3 supply is connected to pin N1 through a 1k resistor. Pin N1 is also connected to a 10k pull-up resistor. A 100nF capacitor is connected between pin N1 and ground. A 10k resistor is connected between pin N1 and ground.
- Diagram 2:** A 3V3 supply is connected to pin N2 through a 1k resistor. Pin N2 is also connected to a 10k pull-up resistor. A 100nF capacitor is connected between pin N2 and ground. A 10k resistor is connected between pin N2 and ground.
- Diagram 3:** A 3V3 supply is connected to pin N3 through a 1k resistor. Pin N3 is also connected to a 10k pull-up resistor. A 100nF capacitor is connected between pin N3 and ground. A 10k resistor is connected between pin N3 and ground.

[illegible]

SSR terminals J4/J9; pin1 = SSR-EN (board +5V, watchdog-gated), pin2 = the switched low side, B03E pins: R7/R20 hold each MOSFET gate down while the B03E pins are high-impedance, N01 opto-isolated - rev B tried that and reverted it; an opto only if the SSR low is powered off-board, and the board powers it.

TC1 terminal J3 / TC2 terminal J8; pin1 = K4, pin2 = K - both logic, biased near AGND only; pin3 = MAX3886 - internal BIAS network. Ungrounded-junction probes required; two grounded-junction probes in one kln would loop through it.

Nav switch J6 is panel-mounted; inputs use ESP32 pull-ups.

J11: IN1/IN2/IN3 (lid/gas-flow/spare) + GND - dry contact each channel to GND.

Display J5 (14-pin, ST7796S + XPT2046 touch module, LCDWIKI MSP4021): 1+ $\rightarrow$ +5V 2 $\rightarrow$ GND 3 $\rightarrow$ CS 4 $\rightarrow$ RS 5 $\rightarrow$ DC 6 $\rightarrow$ SDD/MOSI 7 $\rightarrow$ SDA 8 $\rightarrow$ MISO 10 $\rightarrow$ CLK 11 $\rightarrow$ CS 12 $\rightarrow$ TD 13 $\rightarrow$ TD 14 $\rightarrow$ JRQ. PIN 1 IS +5V, NOT 3V3 – the module regulates on board; do not wire a 3.3V-only panel to it. Logic is 3.3V; touch shares SPI2 with the LCD and both MAX31865 (R39–R43 damping).

J12: CT current inputs, CTA_P/CTA_N/CTB_P/CTB_N – one CT clamp per SSR zone.

J13: DNP SELV voltage sense header. NOT mains-rated, not fitted
- do not wire to AC mains. Y1/C25/C26 load caps are an ASSUME
value (unverified C_L), see design.py comment. ADE7953 I2C
address 0x38 collides with PCF8574A.